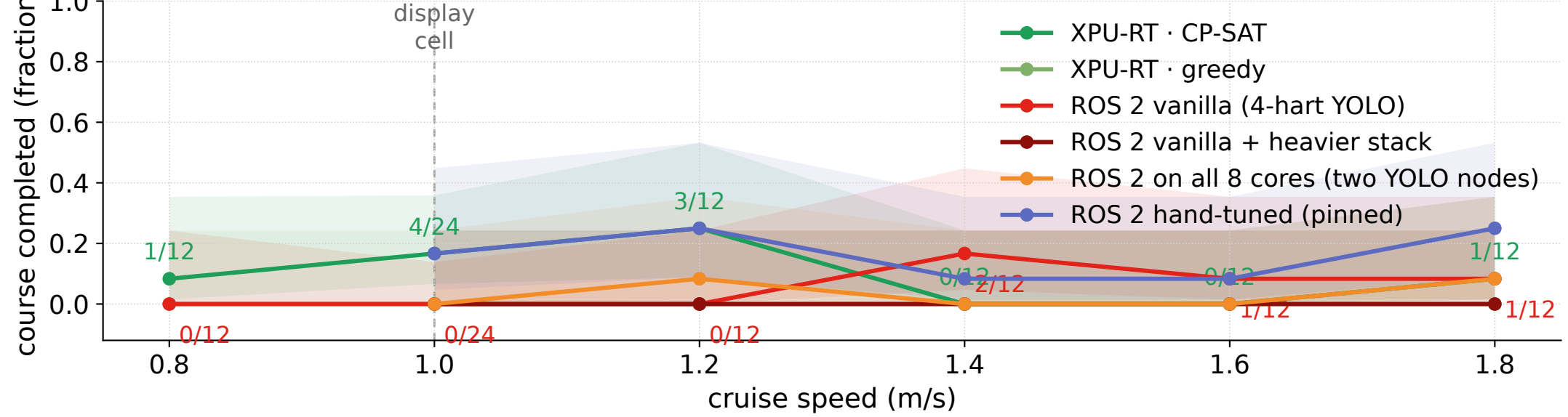
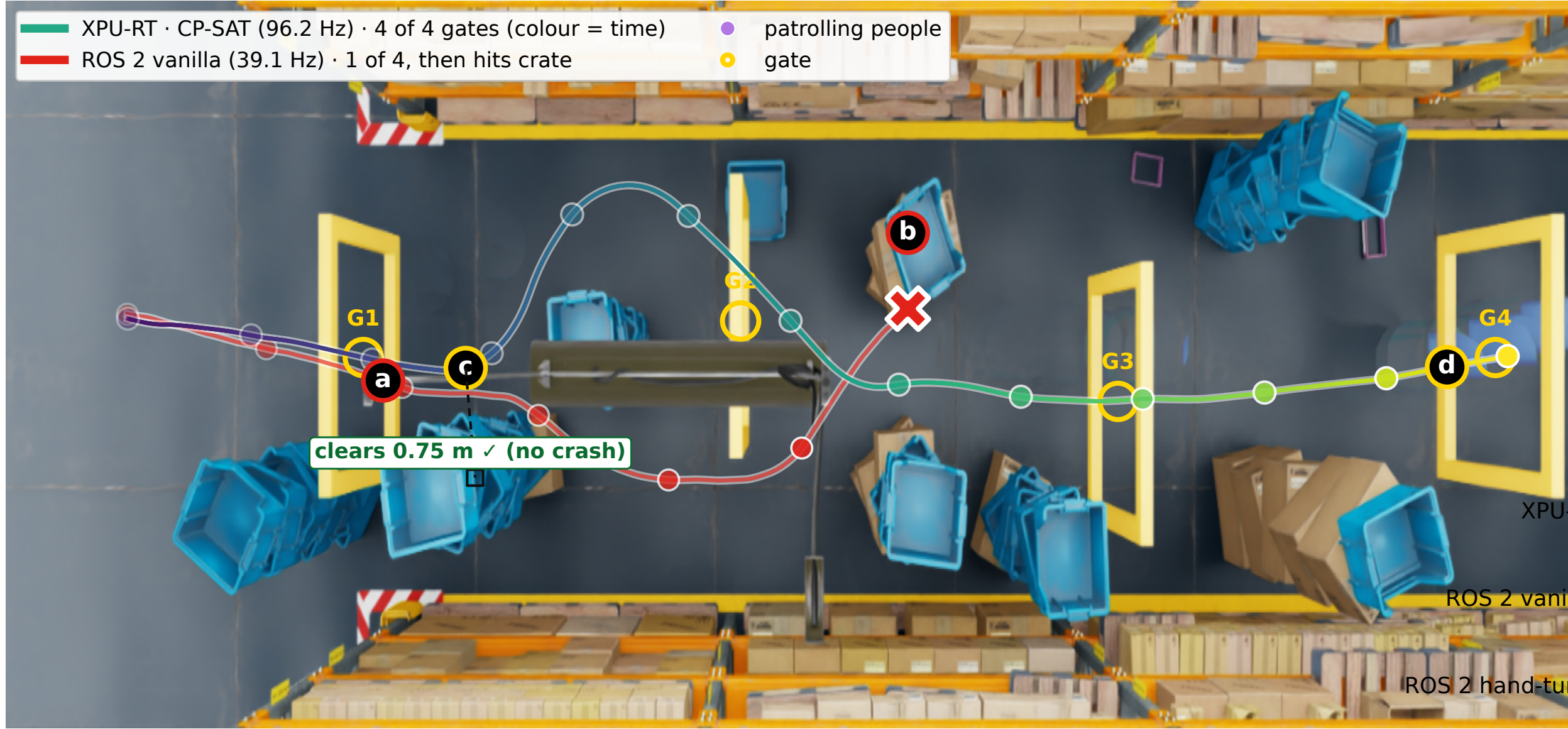


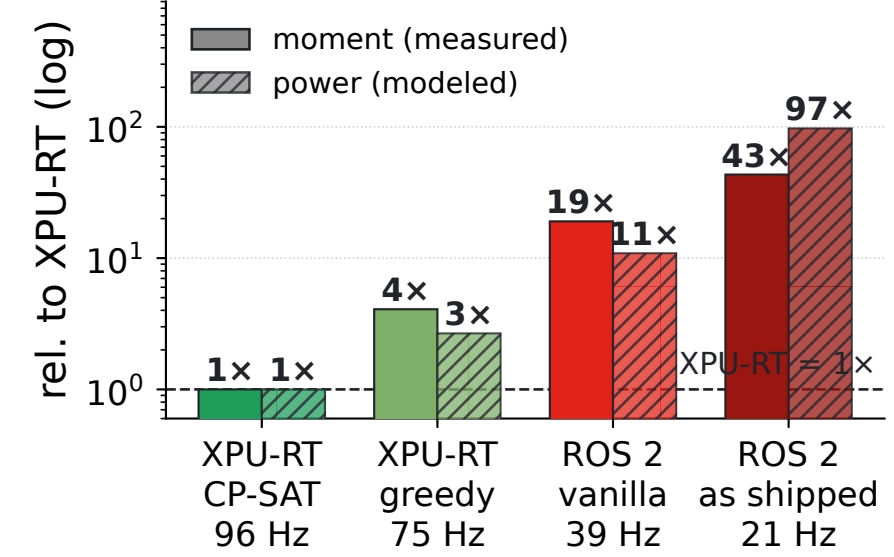
Warehouse gate-course showdown — same scene and controller (1.0 m/s)
only what the K1 delivers differs: control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed
B Flight envelope: success against cruise speed, each arm at its own K1 cadence
only what the K1 delivers differs: control every 10.4 vs 25.5 ms, camera→control latency and goal cadence, replayed



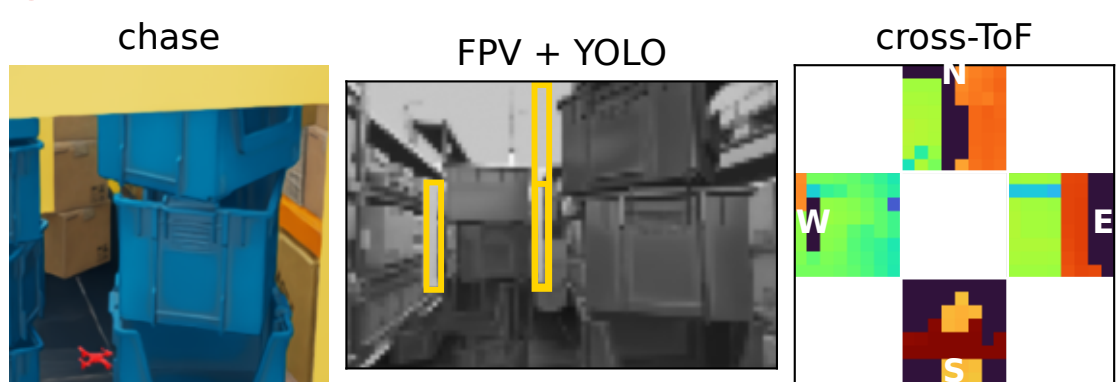
C Generalisation: environments (1.0–1.4 m/s pooled, one flight per seed)

XPU-RT · CP-SAT	8/36	7/48	5/36	6/36	7/36	8/36
ROS 2 vanilla (242 ms)	3/36	2/48	2/36	7/36	5/36	0/36
ROS 2 hand-tuned (56 ms)	7/36	6/36	8/36	7/36	4/24	6/36
	course A d 0.20	course A d 0.30	course A d 0.40	course B d 0.20	course B d 0.30	course B d 0.40

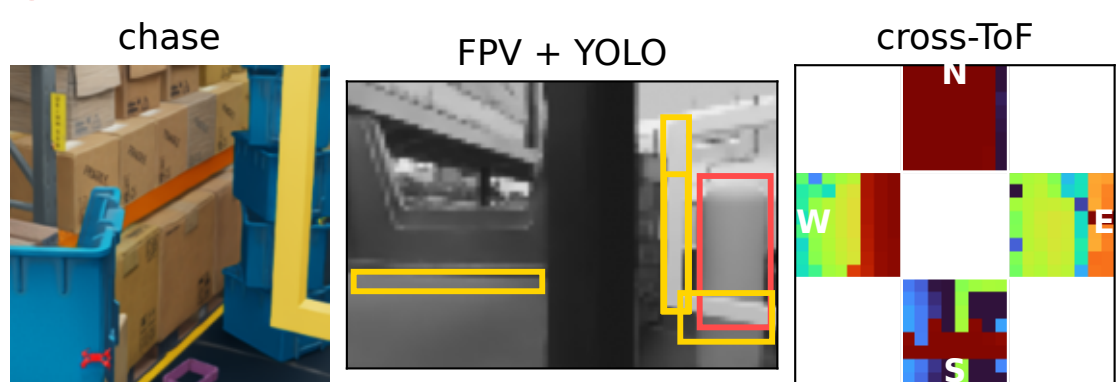
D Mechanism: the baseline thrashes (moment, power)



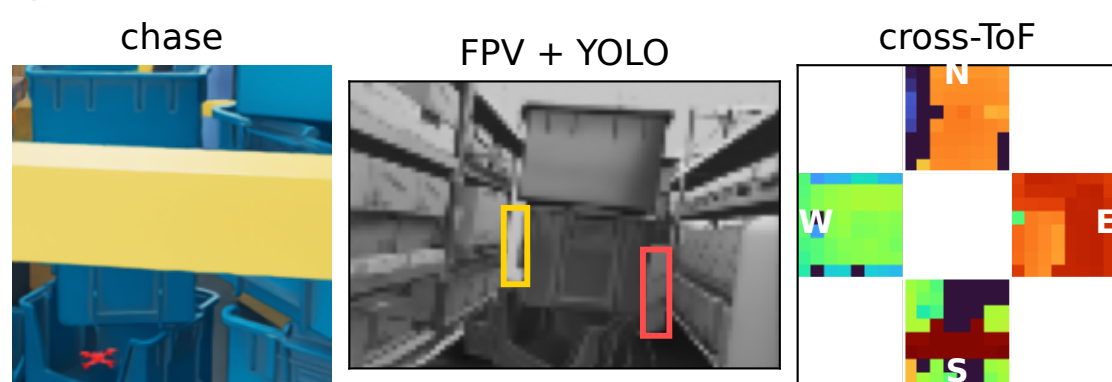
a ROS 2 39 Hz · clears G1 · t=3.5s



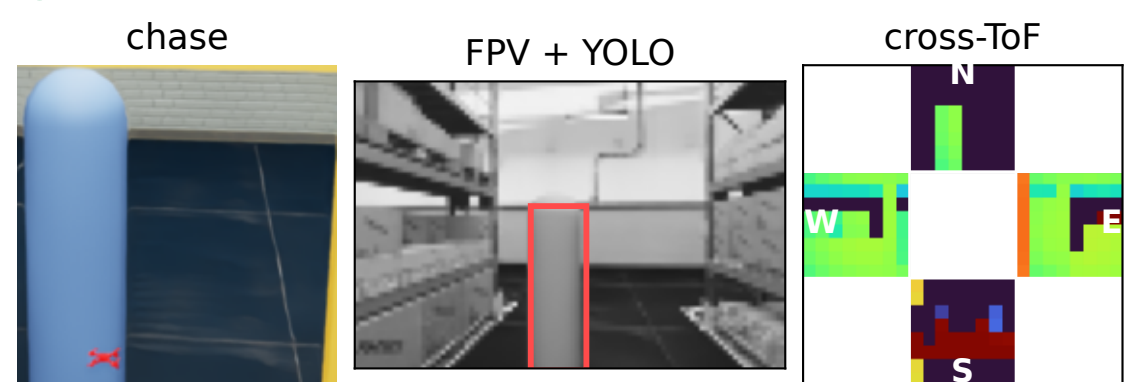
b ROS 2 39 Hz · hits crate after G1 · t=9.5s



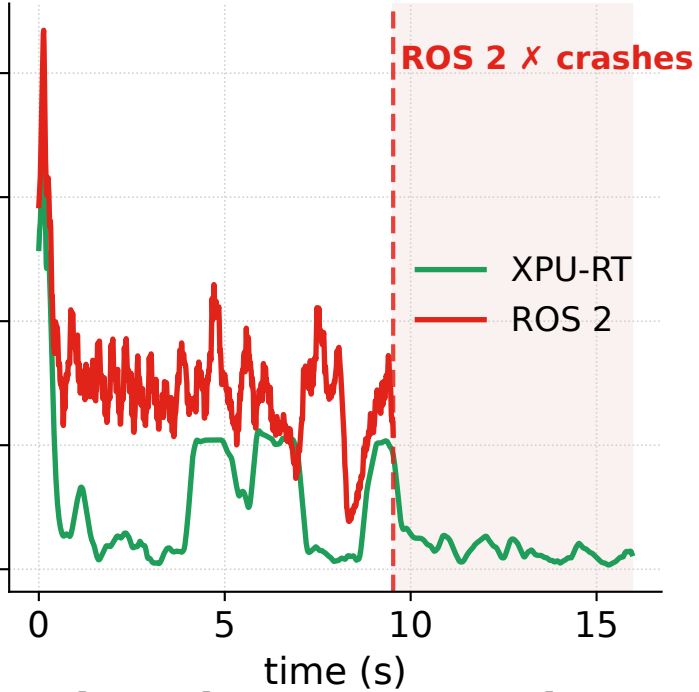
c XPU-RT 96 Hz · clears crate 0.75 m · t=4.3s



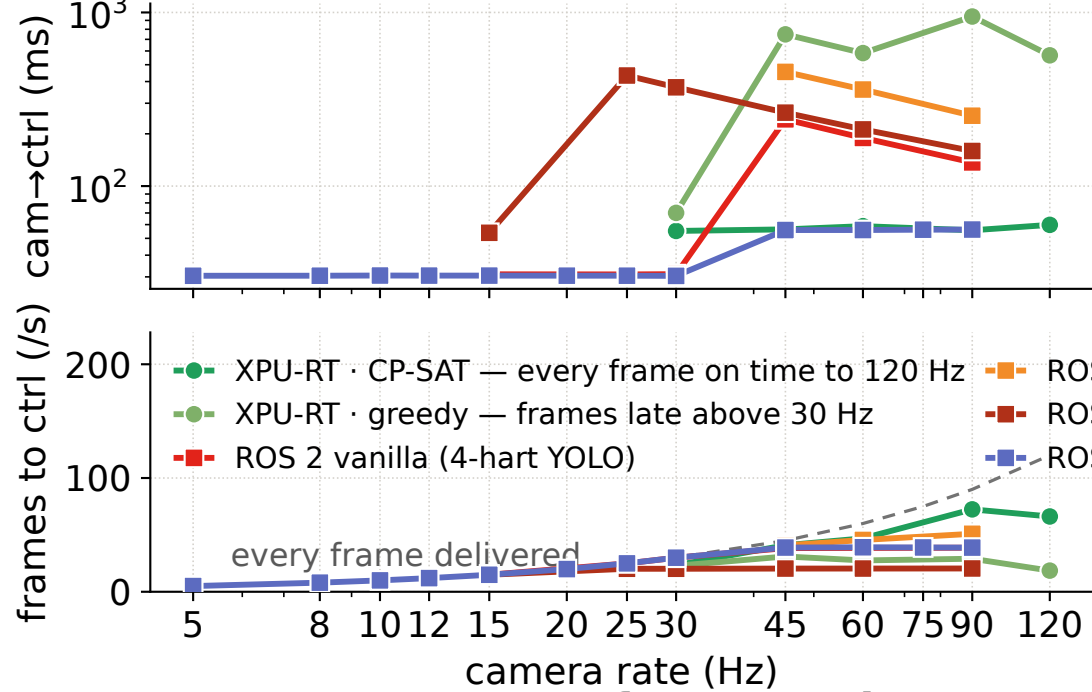
d XPU-RT 96 Hz · reaches gate · t=15.3s



E Body-rate magnitude of the display flights



F Camera rate on the K1: camera→control latency and frames delivered, same kernels

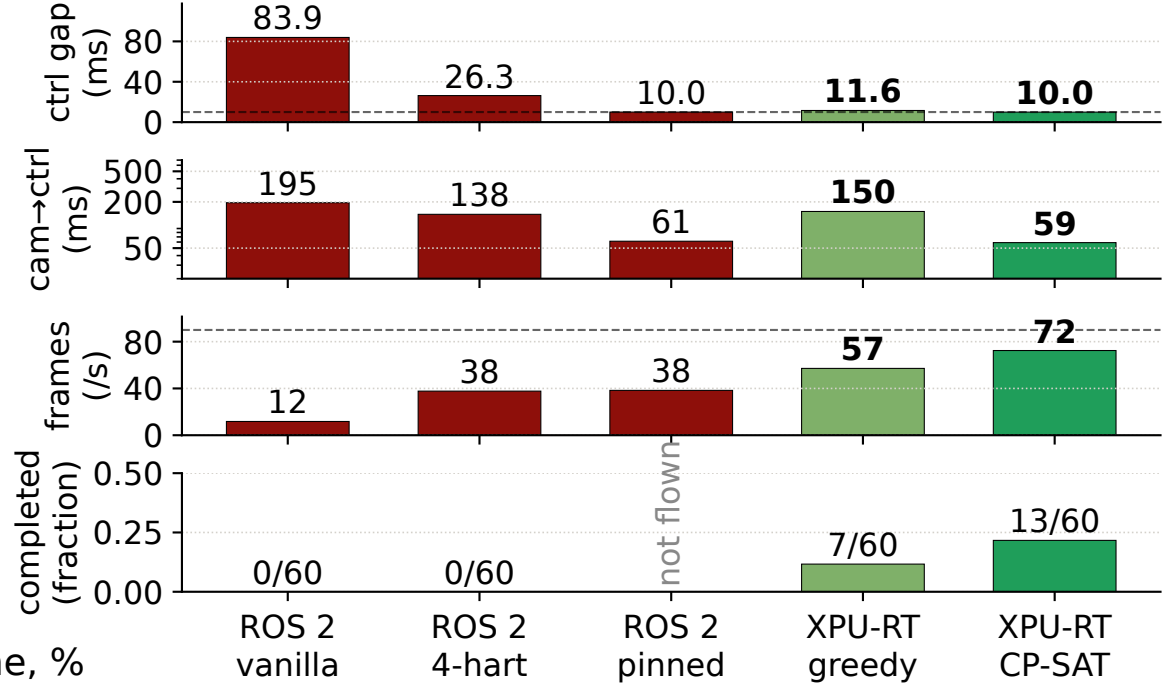


G Where the work lands, 45 Hz camera (board traces, three runs pooled)

XPU-RT CP-SAT 8/8 harts busy	17	9	14	10	17	9	15	8
XPU-RT greedy 5/8 harts busy	32	3	13	2	40	2	8	1
ROS 2 two-YOLO 8/8 harts busy	19	16	16	16	14	6	6	6
ROS 2 vanilla 7/8 harts busy	20	19	19	19	7	6	2	7
	P0	P1	P2	P3	E0	E1	E2	E3

hart (P = performance, E = efficiency cluster) — share of executed inference time, %

H Added load: 90 Hz camera + fnn_block + dronet on the same 8 harts



I Onboard K1, measured, 45 Hz camera — CP-SAT: 57 ms, every frame on time, control 10.0 ms · greedy: 748 ms, 44/44 late, control 13.2 ms
ROS 2 8-core: 447 ms, control 16.6 ms · ROS 2 vanilla: 242 ms, control 25.6 ms

